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NUMERICAL METHODS FOR PARTIAL DIFFERENTIAL EQUATIONS

Fluid-structure interaction

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1 Introduction

In this work, we consider a steady, linear fluid–structure interaction (FSI) problem in both two and three spatial dimensions, involving a viscous incompressible fluid coupled with a linear elastic solid. For clarity of exposition, the mathematical formulation and analysis are presented in the two-dimensional setting; however, all derivations and arguments extend straightforwardly to the three-dimensional case.

The computational domain is decomposed into a fluid subdomain Ω_{fluid} and a solid subdomain Ω_{solid} , separated by the interface Σ . We adopt the *small displacement assumption*, under which the deformation of the solid does not alter the fluid domain. Consequently, the fluid velocity is assumed to vanish on the interface, which simplifies the coupling conditions and the subsequent analysis.

Boundary conditions are prescribed on the remaining portions of the domain, and the fully coupled system provides the basis for the weak formulation and the finite element discretization. The implementation is carried out in C++ using the `deal.II` library, with support for parallel computations. The source code is publicly available at the GitHub repository: https://github.com/DarioSantoroDS/PDE_Project.

2 The problem

$$\begin{cases} -2\nu \nabla \cdot \varepsilon(\mathbf{u}) + \nabla p = 0, & \text{in } \Omega_{\text{fluid}}, \\ \nabla \cdot \mathbf{u} = 0, & \text{in } \Omega_{\text{fluid}}, \\ -\nabla \cdot \sigma(\mathbf{d}) = 0, & \text{in } \Omega_{\text{solid}}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Sigma = \partial\Omega_{\text{fluid}} \cap \partial\Omega_{\text{solid}}, \\ \sigma(\mathbf{d})\mathbf{n} = (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n}, & \text{on } \Sigma. \end{cases} \quad (1)$$

Here, $\varepsilon(\mathbf{u})$ denotes the symmetric gradient of the fluid velocity, defined as

$$\varepsilon(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T),$$

which represents the strain-rate tensor associated with viscous effects in the fluid.

Our goal is to solve the problem (1), which models a two-dimensional steady linear fluid–structure interaction (FSI). The formulation couples the Stokes equations in the fluid region with the linear elasticity equations in the solid region, under the assumption of small displacements. The computational domain has the shape shown in Figure 1 and is partitioned into two subregions, denoted by Ω_{fluid} and Ω_{solid} . Here, $\mathbf{n}$ denotes the unit normal vector on the interface Σ , pointing outward from Ω_{fluid} and inward into Ω_{solid} . The boundary conditions for the overall domain Ω will be discussed later in the subsection. 2.4

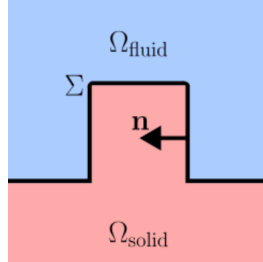


Figure 1: Geometry of the computational domain

2.1 Small displacements assumption

This assumption implies that the deformation of the solid does not modify the domain in which the fluid problem is solved, and that the fluid velocity can be approximated as zero on the interface Σ . In our setting, this means that the fluid velocity vanishes on Σ and that, from the fluid standpoint, the solid geometry remains fixed. As a consequence, the coupling becomes one-directional, which greatly simplifies the analysis developed later in Section 5, although it also makes the model less suitable for realistic applications.

2.2 Symmetric Gradient Formulation for the Fluid

In this work, we adopt the symmetric gradient formulation for the viscous term in the Stokes equations. The momentum balance for an incompressible fluid is written as

$$-2\nu \nabla \cdot \varepsilon(\mathbf{u}) + \nabla p = \mathbf{f} \quad \text{in } \Omega_{\text{fluid}}, \quad (2)$$

where the symmetric gradient is defined as

$$\varepsilon(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T). \quad (3)$$

This formulation naturally arises from the divergence of the viscous stress tensor and is therefore physically consistent with the underlying continuum mechanics model. An alternative formulation replaces the symmetric gradient term with the Laplacian of the velocity, leading to

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{f}. \quad (4)$$

For incompressible flows, i.e. when $\nabla \cdot \mathbf{u} = 0$, and assuming sufficient regularity of the velocity field, the two formulations are equivalent, since

$$-2\nabla \cdot \varepsilon(\mathbf{u}) = -\Delta \mathbf{u}. \quad (5)$$

However, this equivalence relies on the existence of second-order derivatives and on constant viscosity.

For these reasons, and in order to remain consistent with the physical definition of the viscous stress, we employ the symmetric gradient formulation throughout this work.

2.3 Stress Tensor in the Solid

In the solid domain Ω_{solid} , the material is modeled as a linear, isotropic elastic solid. Under the small displacement assumption (Section 2.1), the linearized strain tensor is defined as

$$\varepsilon(\mathbf{d}) = \frac{1}{2}(\nabla \mathbf{d} + (\nabla \mathbf{d})^T), \quad (6)$$

representing the symmetric part of the displacement gradient. The corresponding Cauchy stress tensor is given by Hooke's law:

$$\sigma(\mathbf{d}) = \mathbb{C} : \varepsilon(\mathbf{d}), \quad (7)$$

where $\mathbb{C}$ is the fourth-order elasticity tensor. For an isotropic material, $\mathbb{C}$ depends only on the Lamé constants λ and μ , so that the constitutive relation can be written as

$$\sigma(\mathbf{d}) = 2\mu \varepsilon(\mathbf{d}) + \lambda (\text{tr } \varepsilon(\mathbf{d})) \mathbf{I}. \quad (8)$$

This formulation ensures that the solid satisfies linear momentum balance and accurately describes linear elastic deformations in the small-strain regime.

2.4 Boundary conditions

As shown in Figure 2, let Ω_{fluid} and Ω_{solid} denote the boundaries of the fluid and solid subdomains, respectively. We define $\delta\Omega_{f,1}$ as the upper portion of the fluid boundary, and $\delta\Omega_{f,2} = \delta\Omega_{\text{fluid}} \setminus (\Sigma \cup \delta\Omega_{f,1})$, i.e., the remaining part of the fluid boundary after removing the interface Σ and the upper segment. Similarly, let $\delta\Omega_{s,1} = \delta\Omega_{\text{solid}} \setminus \Sigma$.

The following boundary conditions are imposed:

$$\begin{cases} \mathbf{u} = \mathbf{g}, & \text{on } \delta\Omega_{f,1}, \\ (2\nu \varepsilon(\mathbf{u}) - p \mathbf{I}) \mathbf{n} = \mathbf{0}, & \text{on } \delta\Omega_{f,2}, \\ \mathbf{d} = \mathbf{0}, & \text{on } \delta\Omega_{s,1}. \end{cases} \quad (9)$$

Here:

- On $\delta\Omega_{f,1}$ we prescribe a given fluid velocity $\mathbf{g}$, corresponding to a *Dirichlet boundary condition* for the fluid. This condition may represent either an imposed inflow, an enforced outflow, or a prescribed motion of the fluid, depending on the specific problem setup.
- On $\delta\Omega_{f,2}$ we impose a *natural (Neumann) boundary condition* for the fluid, corresponding to zero traction across this part of the boundary. This condition is commonly used to model stress-free boundaries and may be associated with outflow regions or other free boundaries of the fluid domain.
- On $\delta\Omega_{s,1}$ we enforce zero displacement $\mathbf{d} = \mathbf{0}$, corresponding to a *clamped* or fixed boundary of the solid, where no motion is allowed.

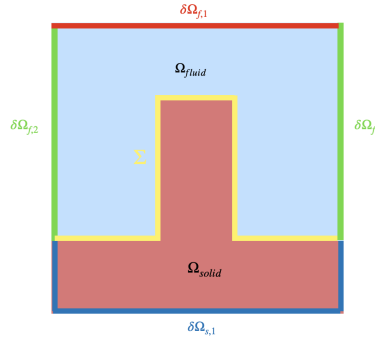


Figure 2: Domain boundaries of the problem.

As for the fluid-solid interface Σ , we enforce a *no-slip* condition for the fluid, i.e., $\mathbf{u} = \mathbf{0}$ on Σ , and we also impose the continuity of the normal stress between the fluid and solid:

$$(2\nu \varepsilon(\mathbf{u}) - p \mathbf{I}) \mathbf{n} = \sigma(\mathbf{d}) \mathbf{n} \quad \text{on } \Sigma. \quad (10)$$

2.5 Complete Problem

The complete fluid–structure interaction problem that we have analyzed and implemented can be stated as follows:

$$\left\{ \begin{array}{ll} -2\nu \nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u}) + \nabla p = 0, & \text{in } \Omega_{\text{fluid}}, \\ \nabla \cdot \mathbf{u} = 0, & \text{in } \Omega_{\text{fluid}}, \\ -\nabla \cdot \boldsymbol{\sigma}(\mathbf{d}) = 0, & \text{in } \Omega_{\text{solid}}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Sigma = \partial\Omega_{\text{fluid}} \cap \partial\Omega_{\text{solid}}, \\ \boldsymbol{\sigma}(\mathbf{d})\mathbf{n} = (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p\mathbf{I})\mathbf{n}, & \text{on } \Sigma, \\ \mathbf{u} = \mathbf{g}, & \text{on } \delta\Omega_{f,1}, \\ (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p\mathbf{I})\mathbf{n} = \mathbf{0}, & \text{on } \delta\Omega_{f,2}, \\ \mathbf{d} = \mathbf{0}, & \text{on } \delta\Omega_{s,1}. \end{array} \right. \quad (11)$$

3 Weak formulation

The weak formulation represents the natural starting point for both the mathematical analysis and the numerical approximation of partial differential equations. By testing the governing equations against suitable functions and integrating by parts, derivatives are transferred from the solution to the test functions, allowing the problem to be posed in Sobolev spaces with weaker regularity requirements.

We start from the fluid momentum equation:

$$-2\nu \nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u}) + \nabla p = 0, \text{ in } \Omega_{\text{fluid}} \quad (12)$$

Multiplying by a test function $\mathbf{v} \in V$, where V is an appropriate function space to be defined later, we obtain the weak form:

$$\int_{\Omega_{\text{fluid}}} (-2\nu \nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u}) + \nabla p) \cdot \mathbf{v} \, d\Omega = 0 \quad (13)$$

Integrating by parts then yields:

$$2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega - \int_{\Omega_{\text{fluid}}} p (\nabla \cdot \mathbf{v}) \, d\Omega = \int_{\partial\Omega_{\text{fluid}}} (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p\mathbf{I})\mathbf{n} \cdot \mathbf{v} \, d\Gamma \quad (14)$$

We observe that the terms associated with Dirichlet boundaries vanish because we impose $\mathbf{v} = \mathbf{0}$ on $\delta\Omega_{f,1}$. Similarly, the terms corresponding to Neumann boundaries are zero as we have homogeneous Neumann conditions. Therefore, we obtain

$$2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega - \int_{\Omega_{\text{fluid}}} p (\nabla \cdot \mathbf{v}) \, d\Omega = 0 \quad (15)$$

Proceeding with the incompressibility condition

$$\nabla \cdot \mathbf{u} = 0, \text{ in } \Omega_{\text{fluid}} \quad (16)$$

we multiply by a test function $q \in Q$, to obtain

$$\int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}) \, d\Omega = 0 \quad (17)$$

Finally, for the solid equilibrium equation

$$-\nabla \cdot \boldsymbol{\sigma}(\mathbf{d}) = 0, \text{ in } \Omega_{\text{solid}} \quad (18)$$

we multiply by a test function $\mathbf{b} \in Z$ and integrate over Ω_{solid} :

$$\int_{\Omega_{\text{solid}}} -(\nabla \cdot \boldsymbol{\sigma}(\mathbf{d})) \cdot \mathbf{b} \, d\Omega = 0 \quad (19)$$

Integrating by parts gives

$$\int_{\Omega_{\text{solid}}} \sigma(\mathbf{d}) : \varepsilon(\mathbf{b}) d\Omega - \int_{\Sigma} (\sigma(\mathbf{d})\mathbf{n}) \cdot \mathbf{b} d\Gamma = 0 \quad (20)$$

On $\partial\Omega_{s,1}$ we set $\mathbf{b} = 0$ due to homogeneous Dirichlet conditions.

On the interface Σ , we use the coupling condition

$$\sigma(\mathbf{d})\mathbf{n} = (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n}, \text{ on } \Sigma \quad (21)$$

leading to

$$\int_{\Omega_{\text{solid}}} \sigma(\mathbf{d}) : \varepsilon(\mathbf{b}) d\Omega - \int_{\Sigma} (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n} \cdot \mathbf{b} d\Gamma = 0 \quad (22)$$

Summing the three contributions for all test functions $\mathbf{a}, q, \mathbf{b}$ gives

$$\begin{aligned} & 2\nu \int_{\Omega_{\text{fluid}}} \varepsilon(\mathbf{u}) : \varepsilon(\mathbf{v}) d\Omega - \int_{\Omega_{\text{fluid}}} p (\nabla \cdot \mathbf{v}) d\Omega + \int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}) d\Omega \\ & + \int_{\Omega_{\text{solid}}} \sigma(\mathbf{d}) : \varepsilon(\mathbf{b}) d\Omega - \int_{\Sigma} (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n} \cdot \mathbf{b} d\Gamma = 0 \end{aligned} \quad (23)$$

This defines a single bilinear form.

3.1 Functional spaces for the coupled problem

The choice of the functional spaces for the unknowns and the test functions follows naturally from the structure of the governing equations and the imposed boundary conditions.

We denote by $\mathbf{V}$ the fluid velocity space, by Q the pressure space, and by $\mathbf{W}$ the solid displacement space.

The fluid velocity is sought in

$$\mathbf{V} := H^1(\Omega_{\text{fluid}})^d,$$

since the viscous term involves first-order derivatives. The associated test space is defined as

$$\mathbf{V}_0 := \{ \mathbf{v} \in H^1(\Omega_{\text{fluid}})^d : \mathbf{v} = \mathbf{0} \text{ on } \delta\Omega_{f,1} \},$$

which incorporates the homogeneous Dirichlet boundary condition on the prescribed portion $\delta\Omega_{f,1}$ of the fluid boundary.

The pressure is sought in

$$Q := L^2(\Omega_{\text{fluid}}),$$

which is the natural space associated with the incompressibility constraint.

The solid displacement is sought in the space

$$\mathbf{W} := \{ \mathbf{w} \in H^1(\Omega_{\text{solid}})^d : \mathbf{w} = \mathbf{0} \text{ on } \delta\Omega_{s,1} \},$$

which already incorporates the homogeneous Dirichlet boundary condition on the fixed part of the solid boundary. Consequently, both the solution and the test functions for the solid belong to the same space $\mathbf{W}$.

3.2 Weak formulation of the coupled problem

The previous considerations define a single bilinear form for the coupled fluid–structure interaction problem. Let

$$y := (\mathbf{u}, p, \mathbf{d}) \in \mathbf{V} \times Q \times \mathbf{W}, \quad z := (\mathbf{v}, q, \mathbf{b}) \in \mathbf{V}_0 \times Q \times \mathbf{W}.$$

We define the bilinear form

$$t : (\mathbf{V} \times Q \times \mathbf{W}) \times (\mathbf{V}_0 \times Q \times \mathbf{W}) \rightarrow \mathbb{R}$$

by

$$\begin{aligned} t(y, z) = & 2\nu \int_{\Omega_{\text{fluid}}} \varepsilon(\mathbf{u}) : \varepsilon(\mathbf{v}) d\Omega - \int_{\Omega_{\text{fluid}}} p (\nabla \cdot \mathbf{v}) d\Omega + \int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}) d\Omega \\ & + \int_{\Omega_{\text{solid}}} \sigma(\mathbf{d}) : \varepsilon(\mathbf{b}) d\Omega - \int_{\Sigma} (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n} \cdot \mathbf{b} d\Gamma. \end{aligned} \quad (24)$$

The weak formulation of the coupled problem then reads: find $y = (\mathbf{u}, p, \mathbf{d}) \in \mathbf{V} \times Q \times \mathbf{W}$ such that

$$t(y, z) = 0 \quad \text{for all } z = (\mathbf{v}, q, \mathbf{b}) \in \mathbf{V}_0 \times Q \times \mathbf{W}. \quad (25)$$

3.3 Decomposition of the bilinear form

For analytical purposes, it is useful to decompose the bilinear form into separate contributions. Let $y = (\mathbf{u}, p, \mathbf{d})$ and $z = (\mathbf{v}, q, \mathbf{b})$. We define:

- Fluid viscous term:

$$a(\mathbf{u}, \mathbf{v}) := 2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega, \quad (26)$$

- Pressure-velocity coupling terms:

$$b(p, \mathbf{v}) := - \int_{\Omega_{\text{fluid}}} (\nabla \cdot \mathbf{v}) p d\Omega, \quad b(q, \mathbf{u}) := \int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}) d\Omega, \quad (27)$$

- Fluid-solid interface coupling:

$$d(\mathbf{u}, p; \mathbf{b}) := - \int_{\Sigma} (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p \mathbf{I}) \mathbf{n} \cdot \mathbf{b} d\Gamma, \quad (28)$$

- Solid elastic term:

$$s(\mathbf{d}, \mathbf{b}) := \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\mathbf{d}) : \boldsymbol{\varepsilon}(\mathbf{b}) d\Omega. \quad (29)$$

Finally, the full bilinear form is given by

$$t(y, z) = a(\mathbf{u}, \mathbf{v}) + b(p, \mathbf{v}) + b(q, \mathbf{u}) + d(\mathbf{u}, p; \mathbf{b}) + s(\mathbf{d}, \mathbf{b}). \quad (30)$$

3.4 Weak formulation with divergence-free Dirichlet lifting

To handle non-homogeneous Dirichlet boundary conditions on $\delta\Omega_{f,1}$, we decompose the velocity as

$$\mathbf{u} = \mathbf{u}_0 + \mathbf{u}_g,$$

where

$$\mathbf{u}_0 \in H_0^1(\Omega_{\text{fluid}})^d \quad \text{with} \quad \mathbf{u}_0 = 0 \text{ on } \delta\Omega_{f,1}$$

vanishes on $\delta\Omega_{f,1}$, and $\mathbf{u}_g \in H^1(\Omega_{\text{fluid}})^d$ is a lifting of the boundary data $\mathbf{g}$, i.e., it satisfies

$$\mathbf{u}_g = \mathbf{g} \quad \text{on } \delta\Omega_{f,1}, \quad \nabla \cdot \mathbf{u}_g = 0 \text{ in } \Omega_{\text{fluid}}.$$

With this choice, the contribution

$$b(q, \mathbf{u}_g) = \int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}_g) d\Omega$$

vanishes identically. Consequently, the lifting $\mathbf{u}_g$ affects only the velocity and fluid-solid interface terms, while the incompressibility constraint involves only the unknown $\mathbf{u}_0$.

The weak formulation in terms of the unknown $\mathbf{u}_0$ reads:

$$\begin{aligned} & 2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}_0) : \boldsymbol{\varepsilon}(\mathbf{a}) d\Omega - \int_{\Omega_{\text{fluid}}} p (\nabla \cdot \mathbf{a}) d\Omega + \int_{\Omega_{\text{fluid}}} q (\nabla \cdot \mathbf{u}_0) d\Omega \\ & + \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\mathbf{d}) : \boldsymbol{\varepsilon}(\mathbf{b}) d\Omega - \int_{\Sigma} (2\nu \boldsymbol{\varepsilon}(\mathbf{u}_0) - p \mathbf{I}) \mathbf{n} \cdot \mathbf{b} d\Gamma \\ & = -2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}_g) : \boldsymbol{\varepsilon}(\mathbf{a}) d\Omega + \int_{\Sigma} 2\nu \boldsymbol{\varepsilon}(\mathbf{u}_g) \mathbf{n} \cdot \mathbf{b} d\Gamma, \\ & \forall (\mathbf{a}, q, \mathbf{b}) \in H_0^1(\Omega_{\text{fluid}})^d \times L^2(\Omega_{\text{fluid}}) \times H_0^1(\Omega_{\text{solid}})^d. \end{aligned} \quad (31)$$

Written in terms of the individual bilinear forms, the system becomes:

$$\begin{aligned} & a(\mathbf{u}_0, \mathbf{a}) + b(p, \mathbf{a}) + b(q, \mathbf{u}_0) + d(\mathbf{u}_0, p; \mathbf{b}) + s(\mathbf{d}, \mathbf{b}) \\ & = -a(\mathbf{u}_g, \mathbf{a}) - d(\mathbf{u}_g, 0; \mathbf{b}), \quad \forall (\mathbf{a}, q, \mathbf{b}) \in Y_0. \end{aligned} \quad (32)$$

Remark: The assumption that the Dirichlet lifting $\mathbf{u}_g$ is divergence-free simplifies the weak formulation by removing its contribution to the incompressibility equation, i.e. $b(q, \mathbf{u}_g) = 0$. This allows the incompressibility constraint to involve only the unknown $\mathbf{u}_0$, while the known lifting $\mathbf{u}_g$ affects only the velocity and interface terms.

4 Galerkin formulation and resulting system

4.1 Finite element spaces

Let $\mathcal{T}_h^f$ and $\mathcal{T}_h^s$ be shape-regular triangulations of Ω_{fluid} and Ω_{solid} , respectively, with mesh parameter $h > 0$. For a given polynomial degree $r \geq 1$, we introduce the space of *continuous piecewise polynomial functions* on a domain D by

$$X_h^r(D) := \left\{ v_h \in C^0(\overline{D}) : v_h|_K \in \mathbb{P}_r(K) \quad \forall K \in \mathcal{T}_h(D) \right\}.$$

To allow for different polynomial degrees for the fluid velocity, pressure, and solid displacement, we define the discrete spaces:

$$\begin{aligned} \mathbf{V}_h &:= (X_h^{r_u}(\Omega_{\text{fluid}}))^d \cap H_0^1(\Omega_{\text{fluid}})^d, & (\text{fluid velocity}) \\ Q_h &:= X_h^{r_p}(\Omega_{\text{fluid}}) \subset L^2(\Omega_{\text{fluid}}), & (\text{fluid pressure}) \\ \mathbf{W}_h &:= (X_h^{r_d}(\Omega_{\text{solid}}))^d \cap H_0^1(\Omega_{\text{solid}})^d, & (\text{solid displacement}) \end{aligned}$$

where $r_u, r_p, r_d \geq 1$ are the chosen polynomial degrees for the velocity, pressure, and displacement, respectively.

The corresponding discrete trial and test space for the coupled problem is then

$$Y_h := \mathbf{V}_h \times Q_h \times \mathbf{W}_h.$$

The Galerkin approximation of the weak formulation with Dirichlet lifting reads: Find $y_h = (\mathbf{u}_{0,h}, p_h, \mathbf{d}_h) \in Y_h$ such that

$$t(y_h, z_h) = f(z_h), \quad \forall z_h = (\mathbf{a}_h, q_h, \mathbf{b}_h) \in Y_h, \quad (33)$$

where $f(z_h)$ collects the contributions of the lifting $\mathbf{u}_g$ as in the continuous case.

Let $\{\phi_i^u\}_{i=1}^{N_u}$, $\{\psi_j\}_{j=1}^{N_p}$, and $\{\phi_k^d\}_{k=1}^{N_d}$ be the finite element basis functions for the spaces $\mathbf{V}_h$, Q_h , and $\mathbf{W}_h$, respectively. We approximate the unknowns as

$$\mathbf{u}_h = \sum_{i=1}^{N_u} U_i \phi_i^u, \quad p_h = \sum_{j=1}^{N_p} P_j \psi_j, \quad \mathbf{d}_h = \sum_{k=1}^{N_d} D_k \phi_k^d, \quad (34)$$

where U_i , P_j , and D_k are the coefficients to determine.

The continuous bilinear forms translate into the following matrices:

- **Fluid viscous term:**

$$a(\mathbf{u}_h, \mathbf{v}_h) \longrightarrow \mathbf{A} \in \mathbb{R}^{N_u \times N_u}, \quad A_{ij} = 2\nu \int_{\Omega_{\text{fluid}}} \varepsilon(\phi_j^u) : \varepsilon(\phi_i^u) d\Omega. \quad (35)$$

- **Pressure-velocity coupling:**

$$b(p_h, \mathbf{v}_h) \longrightarrow \mathbf{B}^T \in \mathbb{R}^{N_p \times N_u}, \quad B_{ij}^T = - \int_{\Omega_{\text{fluid}}} (\nabla \cdot \phi_i^u) \psi_j d\Omega, \quad (36)$$

$$b(q_h, \mathbf{u}_h) \longrightarrow \mathbf{B} \in \mathbb{R}^{N_u \times N_p}. \quad (37)$$

- **Fluid-solid interface coupling:**

$$d(\mathbf{u}_h, p_h; \mathbf{b}_h) \longrightarrow \mathbf{D}_u \in \mathbb{R}^{N_d \times N_u}, \quad \mathbf{D}_p \in \mathbb{R}^{N_d \times N_p}, \quad (38)$$

$$(\mathbf{D}_u)_{ki} = - \int_{\Sigma} 2\nu \varepsilon(\phi_i^u) \mathbf{n} \cdot \phi_k^d d\Gamma, \quad (\mathbf{D}_p)_{kj} = \int_{\Sigma} \psi_j \mathbf{n} \cdot \phi_k^d d\Gamma. \quad (39)$$

- **Solid elastic term:**

$$s(\mathbf{d}_h, \mathbf{b}_h) \longrightarrow \mathbf{S} \in \mathbb{R}^{N_d \times N_d}, \quad S_{kl} = \int_{\Omega_{\text{solid}}} \sigma(\phi_l^d) : \varepsilon(\phi_k^d) d\Omega. \quad (40)$$

4.2 Full discrete system

The full discrete system in block form becomes

$$\begin{bmatrix} \mathbf{A} & \mathbf{B}^T & \mathbf{0} \\ \mathbf{B} & \mathbf{0} & \mathbf{0} \\ \mathbf{D}_u & \mathbf{D}_p & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{P} \\ \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_u \\ \mathbf{F}_p \\ \mathbf{F}_d \end{bmatrix}. \quad (41)$$

For our problem, we have only homogeneous Neumann boundary conditions and a Dirichlet lifting only for the fluid velocity (since $\mathbf{d} = \mathbf{0}$ on $\delta\Omega_{s,1}$). Therefore, the right-hand side vectors for the pressure and solid displacement vanish:

$$\mathbf{F}_p = \mathbf{0}, \quad \mathbf{F}_d = \mathbf{0},$$

while $\mathbf{F}_u$ collects the known contributions from the lifting $\mathbf{u}_g$.

The system then reduces to

$$\begin{bmatrix} \mathbf{A} & \mathbf{B}^T & \mathbf{0} \\ \mathbf{B} & \mathbf{0} & \mathbf{0} \\ \mathbf{D}_u & \mathbf{D}_p & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{P} \\ \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_u \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}. \quad (42)$$

4.3 Block structure of the discrete system

The discrete system derived above is written in terms of three sets of unknowns: the fluid velocity coefficients $\mathbf{U}$, the fluid pressure coefficients $\mathbf{P}$, and the solid displacement coefficients $\mathbf{D}$, leading to a 3×3 block structure at the algebraic level.

From a modeling point of view, the coupled problem can be interpreted as a 2×2 system coupling a fluid subproblem and a solid subproblem. In this interpretation, the fluid operator corresponds to the Stokes system

$$\mathbf{K}_f := \begin{pmatrix} \mathbf{A} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{0} \end{pmatrix}, \quad (43)$$

while the solid operator is given by the elasticity matrix $\mathbf{S}$. The coupling between the two subproblems is induced by the interface matrices $\mathbf{D}_u$ and $\mathbf{D}_p$ arising from the traction continuity condition on Σ .

At this level, the discrete problem admits a recursive 2×2 block structure of the form

$$\begin{pmatrix} \mathbf{K}_f & \mathbf{0} \\ (\mathbf{D}_u \ \mathbf{D}_p) & \mathbf{S} \end{pmatrix}. \quad (44)$$

Here, the upper-left block represents the fluid equations, the lower-right block represents the solid equations, and the off-diagonal block encodes the fluid–solid coupling.

In the present work, this recursive structure is represented in a flattened 3×3 form, where velocity, pressure, and displacement are treated as separate algebraic blocks. This representation is algebraically equivalent to the recursive formulation and preserves the saddle-point structure of the Stokes equations, while remaining compatible with the block matrix framework used in the numerical implementation.

5 Analysis

The first observation is that the fluid part of our problem is completely decoupled from the rest. This allows us to derive results for the fluid independently, and then, under certain assumptions, to study the elastic problem in the solid separately. Such an approach makes it significantly easier to obtain the various analytical results.

5.1 Existence and Uniqueness

5.1.1 Stokes problem in the fluid

Consider the Stokes part of the Galerkin formulation: find $(\mathbf{u}_h, p_h) \in V_h \times Q_h$ such that

$$\begin{cases} a(\mathbf{u}_h, \mathbf{v}_h) + b(\mathbf{v}_h, p_h) = 0, & \forall \mathbf{v}_h \in V_h, \\ b(\mathbf{u}_h, q_h) = 0, & \forall q_h \in Q_h, \end{cases} \quad (45)$$

where

$$a(\mathbf{u}_h, \mathbf{v}_h) = \int_{\Omega_{\text{fluid}}} \nu \nabla \mathbf{u}_h : \nabla \mathbf{v}_h d\Omega, \quad b(\mathbf{v}_h, p_h) = - \int_{\Omega_{\text{fluid}}} p_h (\nabla \cdot \mathbf{v}_h) d\Omega, \quad (46)$$

and the discrete spaces are V_h and Q_h described in section 4.1

By the F. Brezzi theorem, the pair $(\mathbf{u}_h, p_h)$ exists and is unique, provided that the following conditions hold:

1. The bilinear form $a(\cdot, \cdot)$ is coercive on the kernel of b , i.e. there exists $\alpha > 0$ (possibly depending on h) such that

$$a(v_h, v_h) \geq \alpha \|v_h\|_V^2 \quad \forall v_h \in V_h^* := \{v_h \in V_h : b(v_h, q_h) = 0 \quad \forall q_h \in Q_h\}. \quad (47)$$

2. The bilinear forms $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are continuous, i.e. there exist constants $\gamma, \delta > 0$ such that

$$|a(u_h, v_h)| \leq \gamma \|u_h\|_V \|v_h\|_V, \quad |b(v_h, q_h)| \leq \delta \|v_h\|_V \|q_h\|_Q \quad \forall u_h, v_h \in V_h, \quad q_h \in Q_h. \quad (48)$$

3. The discrete inf-sup condition holds: there exists $\beta > 0$ (possibly depending on h) such that

$$\forall q_h \in Q_h, \quad \exists v_h \in V_h : b(v_h, q_h) \geq \beta \|v_h\|_V \|q_h\|_Q. \quad (49)$$

The F. Brezzi theorem is stated in Appendix, section 12.5.

5.1.2 Verification of the F. Brezzi conditions

We now verify that the bilinear forms and discrete spaces introduced above satisfy the hypotheses of the Brezzi (Babuška–Brezzi) theorem. We start by proving the coercivity of the viscous bilinear form $a(\cdot, \cdot)$, which also implies coercivity on the kernel of $b(\cdot, \cdot)$.

Coercivity of the viscous bilinear form

We prove the coercivity of the bilinear form associated with the fluid viscous term. The discrete bilinear form is defined as

$$a(\mathbf{u}_h, \mathbf{v}_h) = 2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}_h) : \boldsymbol{\varepsilon}(\mathbf{v}_h) d\Omega, \quad (50)$$

The velocity space $V_h \subset [H^1(\Omega_{\text{fluid}})]^d$ is endowed with the norm

$$\|\mathbf{v}_h\|_V := \|\nabla \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})}. \quad (51)$$

Thanks to the imposition of homogeneous Dirichlet boundary conditions on a subset of the boundary with positive measure, this seminorm defines a norm on V_h , equivalent to the standard H^1 -norm. In particular, Korn's inequality (see Appendix 12.6) holds on V_h .

Hence, there exists a constant $C_K > 0$, independent of the mesh size h , such that

$$\|\nabla \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})} \leq C_K \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{fluid}})} \quad \forall \mathbf{v}_h \in V_h. \quad (52)$$

As a consequence, for all $\mathbf{v}_h \in V_h$,

$$a(\mathbf{v}_h, \mathbf{v}_h) = 2\nu \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{fluid}})}^2 \geq \frac{2\nu}{C_K^2} \|\nabla \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})}^2 = \alpha \|\mathbf{v}_h\|_V^2, \quad (53)$$

with coercivity constant $\alpha := 2\nu/C_K^2 > 0$.

Since the kernel of the divergence operator, $V_h^* := \{\mathbf{v}_h \in V_h : b(\mathbf{v}_h, q_h) = 0 \quad \forall q_h \in Q_h\}$, is a subspace of V_h , the above estimate holds *a fortiori* for all $\mathbf{v}_h \in V_h^*$. Therefore, the bilinear form $a(\cdot, \cdot)$ is coercive on the kernel of $b(\cdot, \cdot)$, and the first hypothesis of the Brezzi (Babuška–Brezzi) theorem is satisfied.

Continuity of the bilinear forms

We now verify that the bilinear forms $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are continuous with respect to the norms introduced on V_h and Q_h .

Viscous bilinear form $a(\cdot, \cdot)$. By the Cauchy–Schwarz inequality and the definition of the norm $\|\cdot\|_V$, for all $\mathbf{u}_h, \mathbf{v}_h \in V_h$ we have

$$|a(\mathbf{u}_h, \mathbf{v}_h)| = 2\nu \left| \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}_h) : \boldsymbol{\varepsilon}(\mathbf{v}_h) d\Omega \right| \leq 2\nu \|\boldsymbol{\varepsilon}(\mathbf{u}_h)\|_{L^2(\Omega_{\text{fluid}})} \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{fluid}})}. \quad (54)$$

Using Korn’s inequality (see Appendix 12.6), we can bound the symmetric gradient by the full gradient:

$$\|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{fluid}})} \leq \|\nabla \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})} = \|\mathbf{v}_h\|_V. \quad (55)$$

Hence, the bilinear form $a(\cdot, \cdot)$ is continuous:

$$|a(\mathbf{u}_h, \mathbf{v}_h)| \leq \gamma \|\mathbf{u}_h\|_V \|\mathbf{v}_h\|_V, \quad \text{with } \gamma = 2\nu C_K^2. \quad (56)$$

Divergence bilinear form $b(\cdot, \cdot)$. Similarly, for all $\mathbf{v}_h \in V_h$ and $q_h \in Q_h$, we have

$$|b(\mathbf{v}_h, q_h)| = \left| \int_{\Omega_{\text{fluid}}} q_h (\nabla \cdot \mathbf{v}_h) d\Omega \right| \leq \|q_h\|_{L^2(\Omega_{\text{fluid}})} \|\nabla \cdot \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})}. \quad (57)$$

Since $\|\nabla \cdot \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})} \leq \|\nabla \mathbf{v}_h\|_{L^2(\Omega_{\text{fluid}})} = \|\mathbf{v}_h\|_V$, it follows that

$$|b(\mathbf{v}_h, q_h)| \leq \delta \|\mathbf{v}_h\|_V \|q_h\|_Q, \quad \text{with } \delta = 1.$$

Therefore, both bilinear forms are continuous, satisfying the second hypothesis of the Brezzi theorem.

Discrete inf-sup condition. The discrete inf-sup condition (also called the Ladyzhenskaya–Babuška–Brezzi condition) requires that for all $q_h \in Q_h$, there exists $\mathbf{v}_h \in V_h$ such that

$$b(\mathbf{v}_h, q_h) \geq \beta \|\mathbf{v}_h\|_V \|q_h\|_Q,$$

for some constant $\beta > 0$ independent of the mesh size h .

This condition is known to hold for the Taylor–Hood finite element pair (P2 velocity / P1 pressure), which is the choice we adopt in our numerical implementation. Therefore, the third hypothesis of the Brezzi theorem is satisfied.

5.2 Linear elasticity problem in the solid

The discrete linear elasticity problem in the solid, including the fluid-solid coupling via the Neumann term on the interface Σ , reads: find $\mathbf{d}_h \in \mathbf{W}_h$ such that

$$s(\mathbf{d}_h, \mathbf{b}_h) = d(\mathbf{u}_h, p_h; \mathbf{b}_h) \quad \forall \mathbf{b}_h \in \mathbf{W}_h, \quad (58)$$

where we define the linear form

$$d(\mathbf{u}_h, p_h; \mathbf{b}_h) := \int_{\Sigma} \mathbf{t}_h \cdot \mathbf{b}_h d\Gamma, \quad (59)$$

with the traction exerted by the fluid on the interface given by

$$\mathbf{t}_h = (2\nu \boldsymbol{\varepsilon}(\mathbf{u}_h) - p_h \mathbf{I}) \mathbf{n}. \quad (60)$$

The bilinear form $s(\cdot, \cdot)$ corresponds to the standard linear elasticity operator in the solid, and in matrix form reads

$$\mathbf{S} \in \mathbb{R}^{N_d \times N_d}, \quad S_{kl} = \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\boldsymbol{\phi}_l^d) : \boldsymbol{\varepsilon}(\boldsymbol{\phi}_k^d) d\Omega.$$

For the purposes of analysis, we will consider the linear form as a single entity $d(\mathbf{u}_h, p_h; \mathbf{b}_h)$, even though it corresponds in matrix form to the combination of $\mathbf{D}_u$ and $\mathbf{D}_p$:

$$d(\mathbf{u}_h, p_h; \mathbf{b}_h) \longleftrightarrow \mathbf{D}_u \mathbf{u}_h + \mathbf{D}_p p_h.$$

This simplifies the Lax-Milgram analysis while still accounting for the effect of both fluid velocity and pressure on the solid.

In this formulation, the left-hand side represents the solid’s elastic response, whereas the right-hand side corresponds to the fluid traction applied on the interface Σ . Particular attention must be paid to this coupling term, which will be discussed in detail in Section 5.2.1. Existence and uniqueness of the solution $\mathbf{d}_h$ are guaranteed by the Lax-Milgram theorem, provided that $s(\cdot, \cdot)$ is continuous and coercive and that the linear form is continuous. For completeness, the Lax-Milgram theorem is recalled in Appendix 12.3.

5.2.1 Verification of Lax Milgram condtions

We now verify that the bilinear form $s(\cdot, \cdot)$ associated with linear elasticity in the solid satisfies the hypotheses of the Lax-Milgram theorem.

Coercivity. The bilinear form reads

$$s(\mathbf{d}_h, \mathbf{b}_h) = \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\mathbf{d}_h) : \boldsymbol{\varepsilon}(\mathbf{b}_h) d\Omega,$$

where the linear elastic stress is given by

$$\boldsymbol{\sigma}(\mathbf{v}_h) = 2\mu \boldsymbol{\varepsilon}(\mathbf{v}_h) + \lambda (\nabla \cdot \mathbf{v}_h) \mathbf{I},$$

with Lamé parameters $\mu, \lambda > 0$.

Evaluating $s(\mathbf{v}_h, \mathbf{v}_h)$, we get

$$s(\mathbf{v}_h, \mathbf{v}_h) = \int_{\Omega_{\text{solid}}} \left(2\mu \boldsymbol{\varepsilon}(\mathbf{v}_h) : \boldsymbol{\varepsilon}(\mathbf{v}_h) + \lambda (\nabla \cdot \mathbf{v}_h)^2 \right) d\Omega.$$

Since $\mu > 0$ and $\lambda + \frac{2}{3}\mu > 0$ for a physically meaningful material, both terms are non-negative, and in particular

$$s(\mathbf{v}_h, \mathbf{v}_h) \geq 2\mu \int_{\Omega_{\text{solid}}} |\boldsymbol{\varepsilon}(\mathbf{v}_h)|^2 d\Omega = 2\mu \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{solid}})}^2.$$

Defining the norm

$$\|\mathbf{v}_h\|_V := \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{solid}})},$$

we note that this is indeed a norm on $\mathbf{W}_h$, since homogeneous Dirichlet boundary conditions are imposed on a subset of $\partial\Omega_{\text{solid}}$ with positive measure, which rules out rigid body motions. Consequently, $\|\cdot\|_V$ is equivalent to the standard H^1 -norm on $\mathbf{W}_h$. We finally obtain the coercivity estimate

$$s(\mathbf{v}_h, \mathbf{v}_h) \geq \alpha \|\mathbf{v}_h\|_V^2, \quad \text{with } \alpha := 2\mu > 0,$$

which holds for all $\mathbf{v}_h \in \mathbf{W}_h$.

Continuity. The bilinear form reads

$$s(\mathbf{u}_h, \mathbf{v}_h) = \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\mathbf{u}_h) : \boldsymbol{\varepsilon}(\mathbf{v}_h) d\Omega,$$

with

$$\boldsymbol{\sigma}(\mathbf{u}_h) = 2\mu \boldsymbol{\varepsilon}(\mathbf{u}_h) + \lambda (\nabla \cdot \mathbf{u}_h) \mathbf{I}.$$

Applying the Cauchy–Schwarz inequality in $L^2(\Omega_{\text{solid}})$, we get

$$|s(\mathbf{u}_h, \mathbf{v}_h)| \leq \|\boldsymbol{\sigma}(\mathbf{u}_h)\|_{L^2(\Omega_{\text{solid}})} \|\boldsymbol{\varepsilon}(\mathbf{v}_h)\|_{L^2(\Omega_{\text{solid}})}.$$

Next, using the linearity of the stress–strain relation and standard inequalities, we have

$$\|\boldsymbol{\sigma}(\mathbf{u}_h)\|_{L^2(\Omega_{\text{solid}})} \leq (2\mu + d\lambda) \|\boldsymbol{\varepsilon}(\mathbf{u}_h)\|_{L^2(\Omega_{\text{solid}})} = \gamma \|\mathbf{u}_h\|_V,$$

where d is the spatial dimension and $\gamma := 2\mu + d\lambda > 0$.

Combining the two estimates, we obtain

$$|s(\mathbf{u}_h, \mathbf{v}_h)| \leq \gamma \|\mathbf{u}_h\|_V \|\mathbf{v}_h\|_V,$$

which shows that $s(\cdot, \cdot)$ is continuous on $\mathbf{W}_h$.

Continuity of the linear form. The interface traction is defined as

$$\mathbf{t} := (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p \mathbf{I}) \mathbf{n} \quad \text{on } \Sigma. \quad (61)$$

For the continuity of the linear form, we require

1. the fluid velocity satisfies $\mathbf{u} \in [H^1(\Omega_{\text{fluid}})]^d$;
2. the pressure satisfies $p \in L^2(\Omega_{\text{fluid}})$;
3. the interface Σ (and equivalently Ω_{fluid}) is Lipschitz.

These conditions are not mere assumptions: they are satisfied in our setting because the Stokes problem analyzed previously guarantees a solution $(\mathbf{u}_h, p_h)$ with $\mathbf{u}_h \in [H^1(\Omega_{\text{fluid}})]^d$ and $p_h \in L^2(\Omega_{\text{fluid}})$, and the domain Ω_{fluid} is Lipschitz. Therefore, it holds that

$$\mathbf{t} \in [H^{-1/2}(\Sigma)]^d. \quad (62)$$

This regularity ensures that the linear form

$$d(\mathbf{u}_h, p_h; \mathbf{b}_h) = \int_{\Sigma} \mathbf{t}_h \cdot \mathbf{b}_h \, d\Gamma$$

is continuous. Indeed, using the duality between $H^{-1/2}(\Sigma)$ and $H^{1/2}(\Sigma)$, together with the trace inequality

$$\|\mathbf{b}_h\|_{H^{1/2}(\Sigma)} \leq C \|\mathbf{b}_h\|_V,$$

we obtain

$$|d(\mathbf{u}_h, p_h; \mathbf{b}_h)| \leq \|\mathbf{t}_h\|_{H^{-1/2}(\Sigma)} \|\mathbf{b}_h\|_{H^{1/2}(\Sigma)} \leq C \|\mathbf{t}_h\|_{H^{-1/2}(\Sigma)} \|\mathbf{b}_h\|_V,$$

for some constant $C > 0$ independent of the mesh size h . Hence, the linear form is continuous.

5.2.2 Conclusion

Under the assumptions and results discussed above, we have verified that:

- the Stokes problem in the fluid satisfies all the hypotheses of the Brezzi (Babuška–Brezzi) theorem: coercivity of the viscous bilinear form on the kernel of the divergence, continuity of the bilinear forms, and the discrete inf-sup condition;
- the linear elasticity problem in the solid satisfies the conditions of the Lax–Milgram theorem: the bilinear form $s(\cdot, \cdot)$ is continuous and coercive on $\mathbf{W}_h$, and the right-hand side linear form $d(\mathbf{u}_h, p_h; \cdot)$ is continuous.

Therefore, the coupled fluid–solid problem admits a unique solution $(\mathbf{u}_h, p_h, \mathbf{d}_h) \in V_h \times Q_h \times \mathbf{W}_h$, and the Galerkin formulation is well-posed.

5.3 Stability

In addition to existence and uniqueness, we can also derive stability estimates for the coupled fluid–solid problem.

Fluid part. From the F. Brezzi theorem (see Appendix 12.5), the discrete solution $(\mathbf{u}_h, p_h) \in V_h \times Q_h$ of the Stokes problem satisfies the following a-priori estimates:

$$\|\mathbf{u}_h\|_V \leq \frac{1}{\alpha} \|f\|_{V'}, \quad \|p_h\|_Q \leq \frac{1}{\beta} \left(1 + \frac{\gamma}{\alpha}\right) \|f\|_{V'}, \quad (63)$$

where α is the coercivity constant of the viscous bilinear form, β is the discrete inf-sup constant, γ is the continuity constant of $a(\cdot, \cdot)$, and $f \in V'$ represents the right-hand side forcing.

Solid part. Similarly, by the Lax–Milgram theorem (see Appendix 12.3), the solution $\mathbf{d}_h \in \mathbf{W}_h$ of the linear elasticity problem with the interface traction as right-hand side satisfies

$$\|\mathbf{d}_h\|_V \leq \frac{1}{\alpha_s} \|d(\mathbf{u}_h, p_h; \cdot)\|_{V'}, \quad (64)$$

where α_s is the coercivity constant of the bilinear form $s(\cdot, \cdot)$ and $\|d(\mathbf{u}_h, p_h; \cdot)\|_{V'}$ is the dual norm of the linear form representing the fluid traction.

Coupled stability. The stability estimate for the solid displacement involves the linear form $d(\mathbf{u}_h, p_h; \cdot)$, which depends on the fluid velocity and pressure. However, this dependence does not compromise the overall stability of the coupled problem.

Indeed, from the continuity of the linear form established in Section 5.2.1, we have

$$\|d(\mathbf{u}_h, p_h; \cdot)\|_{V'} \leq C (\|\mathbf{u}_h\|_V + \|p_h\|_Q),$$

for some constant $C > 0$ independent of the mesh size h .

Combining this estimate with the stability bounds for the Stokes problem,

$$\|\mathbf{u}_h\|_V + \|p_h\|_Q \leq C_f \|f\|_{V'},$$

we obtain

$$\|\mathbf{d}_h\|_V \leq C_s \|f\|_{V'},$$

where $C_s > 0$ is a constant independent of h .

Therefore, the coupled fluid–solid problem satisfies the global stability estimate

$$\|\mathbf{u}_h\|_V + \|p_h\|_Q + \|\mathbf{d}_h\|_V \leq C \|f\|_{V'}, \quad (65)$$

with a constant $C > 0$ independent of the mesh size h . This shows that the solution depends continuously on the problem data.

5.4 Convergence estimates

For the fluid part, using Taylor–Hood finite elements of order $k + 1$ for the velocity and order k for the pressure ($P(k + 1)/Pk$), the Brezzi theory together with standard interpolation estimates provides the following convergence result:

$$\|\mathbf{u} - \mathbf{u}_h\|_V + \|p - p_h\|_Q \leq C h^k (\|\mathbf{u}\|_{H^{k+1}(\Omega_{\text{fluid}})} + \|p\|_{H^k(\Omega_{\text{fluid}})}), \quad (66)$$

where h is the mesh size and $C > 0$ is independent of h . This estimate reflects the classical optimal convergence rates for the chosen Taylor–Hood pair.

For the solid part, the right-hand side depends on the fluid solution through the interface traction. Applying Céa’s lemma (Appendix 12.8) yields

$$\|\mathbf{d} - \mathbf{d}_h\|_V \leq C \left(\inf_{\mathbf{b}_h \in \mathbf{W}_h} \|\mathbf{d} - \mathbf{b}_h\|_V + \|d(\mathbf{u}, p; \cdot) - d(\mathbf{u}_h, p_h; \cdot)\|_{V'} \right), \quad (67)$$

where $V' = \mathbf{W}'_h$ denotes the dual space.

Using standard interpolation estimates for finite elements of order r , we have

$$\inf_{\mathbf{b}_h \in \mathbf{W}_h} \|\mathbf{d} - \mathbf{b}_h\|_V \leq C h^r \|\mathbf{d}\|_{H^{r+1}(\Omega_{\text{solid}})}.$$

Moreover, by continuity of the interface linear form and the trace operator,

$$\|d(\mathbf{u}, p; \cdot) - d(\mathbf{u}_h, p_h; \cdot)\|_{V'} \leq C (\|\mathbf{u} - \mathbf{u}_h\|_V + \|p - p_h\|_Q).$$

Combining these estimates with the convergence result for the Stokes problem, we obtain

$$\|\mathbf{d} - \mathbf{d}_h\|_V \leq C (h^r \|\mathbf{d}\|_{H^{r+1}(\Omega_{\text{solid}})} + h^k (\|\mathbf{u}\|_{H^{k+1}(\Omega_{\text{fluid}})} + \|p\|_{H^k(\Omega_{\text{fluid}})})), \quad (68)$$

with a constant $C > 0$ independent of h .

Hence, depending on the polynomial orders selected for the velocity, pressure, and displacement fields, different convergence rates are achieved for the fluid and solid components. In the coupled setting considered here, the convergence of the solid displacement is also influenced by the accuracy with which the fluid traction on the interface is approximated.

As a consequence of the estimate above, the convergence rate of the solid displacement is governed by the slower of the two error contributions, namely the solid discretization error of order h^r and the fluid-induced error of order h^k . Therefore, the overall convergence rate for $\mathbf{d}_h$ is of order $h^{\min(r, k)}$.

This observation indicates that choosing a polynomial degree $r > k$ for the solid discretization does not improve the asymptotic convergence rate of the coupled problem, since the error is then dominated by the fluid approximation. An optimal balance is achieved by selecting comparable approximation orders for the fluid and solid subproblems.

6 Implementation

6.1 Sezione provvisoria solo per contesto

We consider a stationary fluid–structure interaction problem in which different physical models are posed on disjoint subdomains of a common computational domain Ω . The domain is decomposed as

$$\Omega = \Omega_{\text{fluid}} \cup \Omega_{\text{solid}}, \quad \Omega_{\text{fluid}} \cap \Omega_{\text{solid}} = \emptyset,$$

where Ω_{fluid} denotes the fluid subdomain and Ω_{solid} the solid subdomain. The two regions are separated by the interface

$$\Sigma = \partial\Omega_{\text{fluid}} \cap \partial\Omega_{\text{solid}}.$$

Fluid model. In the fluid region Ω_{fluid} , we consider the stationary incompressible Stokes equations written in strain–tensor form. The unknowns are the fluid velocity u and the pressure p , satisfying

$$-2\nu\nabla \cdot \varepsilon(u) + \nabla p = 0 \quad \text{in } \Omega_{\text{fluid}},$$

$$\nabla \cdot u = 0 \quad \text{in } \Omega_{\text{fluid}},$$

where ν denotes the kinematic viscosity and

$$\varepsilon(u) = \frac{1}{2} (\nabla u + \nabla u^T)$$

is the symmetric gradient of the fluid velocity.

On a portion of the external boundary of the fluid domain, $\partial\Omega_{f,1} \subset \partial\Omega \cap \partial\Omega_{\text{fluid}}$, a Dirichlet boundary condition is prescribed,

$$u = g \quad \text{on } \partial\Omega_{f,1},$$

while on the remaining part of the fluid boundary, $\partial\Omega_{f,2}$, a natural (free–flow) condition is imposed,

$$(2\nu\varepsilon(u) - pI)n = 0 \quad \text{on } \partial\Omega_{f,2}.$$

Solid model. The solid occupies the subdomain Ω_{solid} . Its deformation is described by the displacement field d , which satisfies the equilibrium equations

$$-\nabla \cdot \sigma(d) = 0 \quad \text{in } \Omega_{\text{solid}},$$

where $\sigma(d)$ denotes the Cauchy stress tensor associated with the solid displacement. In the present setting, the solid is assumed to undergo small deformations.

On the external boundary of the solid, $\partial\Omega_{s,1} \subset \partial\Omega \cap \partial\Omega_{\text{solid}}$, the solid is clamped,

$$d = 0 \quad \text{on } \partial\Omega_{s,1}.$$

Interface conditions. Under the assumption of small deformations, the coupling between fluid and solid is one–way: the fluid exerts forces on the solid, while the deformation of the solid does not affect the fluid domain. On the interface Σ , the following conditions are imposed.

First, a no–slip condition for the fluid velocity,

$$u = 0 \quad \text{on } \Sigma.$$

Second, continuity of normal tractions, stating that the stress acting on the solid equals the normal stress exerted by the fluid,

$$\sigma(d)n = (2\nu\varepsilon(u) - pI)n \quad \text{on } \Sigma,$$

where n denotes the unit normal vector on Σ pointing from the solid into the fluid.

Weak formulation. The weak formulation is obtained by multiplying the governing equations by suitable test functions and integrating by parts over the corresponding subdomains. We seek

$$y = \{u, p, d\} \in Y$$

such that

$$2\nu(\varepsilon(u), \varepsilon(v))_{\Omega_{\text{fluid}}} - (\nabla \cdot v, p)_{\Omega_{\text{fluid}}} - (q, \nabla \cdot u)_{\Omega_{\text{fluid}}} + (\sigma(d), \varepsilon(b))_{\Omega_{\text{solid}}} - (b, (2\nu\varepsilon(u) - pI)n)_{\Sigma} = 0$$

for all admissible test functions (v, q, b) belonging to the corresponding test space Y .

6.2 Extended variables and discrete spaces

Before discussing implementation details, it is important to observe that the coupled problem involves multiple solution variables. In particular, the unknowns are the fluid velocity u , the fluid pressure p , and the solid displacement d . Within `deal.II`, such problems are naturally formulated as vector-valued problems, and the general design principles are discussed in the documentation module on handling vector-valued problems.

Extension of variables to the whole domain. In the continuous formulation, the fluid variables u and p are only defined on the fluid subdomain Ω_{fluid} . For implementation purposes, however, it is convenient to extend these variables by zero to the entire domain Ω . In general, this extension is discontinuous across the interface Σ . On Ω_{fluid} we require

$$u \in H^1(\Omega_{\text{fluid}})^d, \quad p \in L^2(\Omega_{\text{fluid}}),$$

and for their extensions to Ω we introduce the spaces

$$u \in V = \{\tilde{u} \mid \tilde{u}|_{\Omega_{\text{fluid}}} \in H^1(\Omega_{\text{fluid}})^d, \tilde{u}|_{\Omega_{\text{solid}}} = 0\},$$

$$p \in P = \{\tilde{p} \mid \tilde{p}|_{\Omega_{\text{fluid}}} \in L^2(\Omega_{\text{fluid}}), \tilde{p}|_{\Omega_{\text{solid}}} = 0\}.$$

For simplicity, boundary conditions are omitted from the definition of these spaces, since they are not essential for the present discussion. In what follows, the tilde notation is dropped, and the same symbols u and p are used to denote both the original and the extended variables.

Discrete fluid spaces and zero extensions. For the Stokes problem, a standard stable choice on the fluid subdomain is the Taylor–Hood pair $Q_{p+1}^d \times Q_p$. Since this choice is only meaningful on Ω_{fluid} , we define discrete spaces on the full triangulation $\mathcal{T}$ by extending the fields by zero on cells belonging to the solid region:

$$V_h = \left\{ u_h \left| \begin{array}{l} \forall K \in \mathcal{T} : u_h|_K \in Q_{p+1}^d \text{ if } K \subset \Omega_{\text{fluid}}, \\ u_h|_{\Omega_{\text{fluid}}} \text{ continuous,} \\ u_h|_K = 0 \text{ if } K \subset \Omega_{\text{solid}} \end{array} \right. \right\} \subset V,$$

$$P_h = \left\{ p_h \left| \begin{array}{l} \forall K \in \mathcal{T} : p_h|_K \in Q_p \text{ if } K \subset \Omega_{\text{fluid}}, \\ p_h|_{\Omega_{\text{fluid}}} \text{ continuous,} \\ p_h|_K = 0 \text{ if } K \subset \Omega_{\text{solid}} \end{array} \right. \right\} \subset P.$$

On the solid cells, the requirement that the discrete fields vanish identically leads to the need for a finite element space with no degrees of freedom. This is provided in `deal.II` by the `FE_Nothing` class, which represents a zero-dimensional finite element. Its defining property is that it contributes no degrees of freedom, i.e.,

$$\text{FiniteElement}::\text{n_dofs_per_cell}() = 0.$$

For clarity, we denote this space by

$$Z = \{\varphi : \varphi(x) = 0\},$$

and correspondingly choose $Z_h = Z$.

Solid variables and discrete spaces. The same construction applies to the solid displacement. The displacement field d is physically defined only on Ω_{solid} , and we extend it by zero to the fluid region. The corresponding function space is

$$d \in U = \{ \tilde{d} \mid \tilde{d}|_{\Omega_{\text{solid}}} \in H^1(\Omega_{\text{solid}})^d, \tilde{d}|_{\Omega_{\text{fluid}}} \in Z^d \}.$$

A typical discrete choice of polynomial degree r is

$$U_h = \left\{ d_h \left| \begin{array}{l} \forall K \in \mathcal{T} : d_h|_K \in Q_r^d \text{ if } K \subset \Omega_{\text{solid}}, \\ d_h|_{\Omega_{\text{solid}}} \text{ continuous,} \\ d_h|_K \in Z^d \text{ if } K \subset \Omega_{\text{fluid}} \end{array} \right. \right\} \subset U.$$

Global discrete solution space. Collecting all components, the discrete solution is the vector-valued field

$$y_h = \{u_h, p_h, d_h\},$$

which belongs to the space

$$Y_h = \left\{ y_h = \{u_h, p_h, d_h\} : \begin{array}{l} y_h|_{\Omega_{\text{fluid}}} \in Q_{p+1}^d \times Q_p \times Z^d, \\ y_h|_{\Omega_{\text{solid}}} \in Z^d \times Z \times Q_r^d \end{array} \right\}.$$

This construction allows the use of a single global triangulation and a single `DoFHandler`, while correctly representing variables that are active only on one of the two subdomains.

Mapping the extended formulation to the hp framework. The extension-by-zero argument explains the structure of the discrete space Y_h , but its practical realization relies on the ability to associate different finite element spaces with different cells of the same triangulation. This is achieved using the hp finite element framework, which allows different cells to employ different local discretizations while sharing a common vector-valued structure.

In the present setting, fluid cells use the finite element $Q_{p+1}^d \times Q_p \times Z^d$, while solid cells use $Z^d \times Z \times Q_r^d$. Both elements have the same number of vector components, but differ in their local polynomial content, with identically zero components represented through the space Z . The hp framework provides the necessary infrastructure to assign these elements to cells according to their subdomain affiliation and to manage the resulting degrees of freedom within a single global discretization.

This approach handles the bookkeeping aspects of the discretization—namely, the restriction of variables to their respective subdomains and the presence of identically zero components—in a uniform and systematic way. The treatment of coupling effects across the interface, as well as the enforcement of interface conditions, is addressed separately and is discussed in the following subsections.

6.3 Identifying the subdomain of each cell

The implementation of the coupled formulation requires a reliable mechanism to identify, for each mesh cell, whether it belongs to the fluid or to the solid subdomain. This information is essential for assembling the correct volume contributions, detecting interface faces, enforcing interface conditions, and selecting the appropriate finite element discretization on each cell.

To this end, each cell of the mesh is assigned an identifier of the *physical subdomain* at mesh generation time, indicating whether it belongs to Ω_{fluid} or Ω_{solid} . This identifier is inherited by child cells under mesh refinement, ensuring that the physical subdomain classification remains consistent throughout adaptive refinement cycles. As a result, the subdomain to which a given active cell belongs can be determined unambiguously at any stage of the computation.

The physical subdomain identifier is then used to select the appropriate local finite element discretization. Cells belonging to the fluid subdomain are associated with a finite element suitable for the Stokes equations, while cells in the solid subdomain are assigned a finite element for linear elasticity. This choice is updated at the beginning of each refinement cycle to reflect the current mesh configuration. In this way, different physical models are discretized on different parts of the domain within a single, unified computational framework.

6.4 Mesh refinement

An additional challenge in the present implementation is the construction of a suitable mesh refinement strategy. Since the goal is to rely on an error estimator that is robust and broadly applicable, a residual-based estimator of

Kelly type is employed. Such estimators are widely used and inexpensive, and they can be applied to a large class of problems without requiring problem-specific modifications.

In the coupled setting considered here, error indicators are computed separately for the fluid and solid variables. More precisely, the estimator is applied to the discrete solution y_h by considering only the velocity components on fluid cells and only the displacement components on solid cells. This results in two sets of cell-wise error indicators, one associated with the Stokes equations and one associated with the elasticity equations. Since these quantities have different physical units and may differ by several orders of magnitude, the corresponding indicators are normalized before being combined into a single refinement indicator that drives mesh adaptation.

To understand the behavior of this refinement strategy, it is useful to recall the structure of the Kelly error estimator. For a scalar finite element field u_h , the local error indicator associated with a cell K is given by

$$\eta_K^2 = \sum_{F \subset \partial K} c_F \int_F [[\nabla u_h \cdot n]]^2 ds,$$

where the sum runs over all faces F of the cell, $[[\cdot]]$ denotes the jump of the enclosed quantity across the face, n is a unit normal vector, and c_F is a face-dependent scaling factor. The global error estimator is then defined as

$$\eta^2 = \sum_K \eta_K^2,$$

and satisfies $\eta \approx \|\nabla(u - u_h)\|$ for standard elliptic problems. In the vector-valued case, the estimator is applied component-wise and combined according to the selected component mask.

In the present multiphysics formulation, the discrete solution is extended by zero outside its natural subdomain. As a consequence, across faces belonging to the interface Σ between the fluid and solid regions, the discrete field or its gradient exhibits an artificial jump. These jumps are not related to the discretization error but arise solely from the zero extension across the interface. Crucially, their magnitude does not decrease under mesh refinement.

As a result, the Kelly estimator produces disproportionately large error indicators in a layer of cells adjacent to the interface Σ on both sides. The refinement process therefore focuses almost exclusively on this region, even when the discretization error away from the interface would warrant refinement elsewhere. This behavior is not a defect of the discretization itself, but rather a limitation of using a generic jump-based estimator that is unaware of the multiphysics nature of the problem and interprets the interface as a legitimate location for large residual contributions.

Ideally, an error estimator tailored to the coupled problem would explicitly exclude interface faces when evaluating jump terms. In the absence of such a specialized estimator, a pragmatic heuristic is adopted: the error indicators of cells adjacent to the interface Σ are set to zero before the refinement step. This suppresses the artificial spikes in the error indicator and leads to a more balanced refinement of the mesh.

Although this strategy might appear to prevent refinement near the interface, this is not the case in practice. The restriction that neighboring cells may differ by at most one refinement level ensures that cells close to the interface Σ are still refined indirectly when adjacent regions require refinement. While this approach is clearly suboptimal from a theoretical standpoint, it proves effective in practice and keeps the focus of mesh refinement on regions where the discretization error is genuinely large, while leaving room for future improvements based on more problem-aware error estimators.

6.5 Dealing with the interface terms

We now address the treatment of the interface contributions in the discrete weak formulation of the coupled problem. At the discrete level, the Galerkin formulation reads: find

$$y_h = \{u_h, p_h, d_h\} \in Y_h$$

such that

$$2\nu(\varepsilon(v_h), \varepsilon(u_h))_{\Omega_{\text{fluid}}} - (\nabla \cdot v_h, p_h)_{\Omega_{\text{fluid}}} - (q_h, \nabla \cdot u_h)_{\Omega_{\text{fluid}}} + (\sigma(d_h), \varepsilon(b_h))_{\Omega_{\text{solid}}} - (b_h, (2\nu\varepsilon(u_h) - p_h I)n)_{\Sigma} = 0$$

for all admissible test functions $\{v_h, q_h, b_h\} \in Y_h$.

Although all discrete variables are extended by zero outside their respective subdomains, the assembly is carried out by distinguishing between fluid and solid cells and adding only the contributions associated with the physical model posed on each cell. The volume terms in Ω_{fluid} and Ω_{solid} are assembled independently using standard procedures for the Stokes and linear elasticity problems.

Interface contribution. The interface term

$$-(b_h, (2\nu\varepsilon(u_h) - p_h I)n)_\Sigma$$

requires special treatment. Assuming that the interface Σ coincides with element faces, it can be written as a sum of integrals over interface faces. Denoting by $\psi_i \in Y_h$ the global shape functions and by $\psi_i[\cdot]$ their components, the contribution to a matrix entry (i, j) takes the form

$$-\sum_K (\psi_i[d], (2\nu\varepsilon(\psi_j[u]) - \psi_j[p]I)n)_{\partial K \cap \Sigma}.$$

The quantities involved in this term are naturally defined on different sides of the interface: the displacement test function $\psi_i[d]$ and the normal vector n are associated with the solid side, while the fluid stress depends on velocity and pressure values taken from the fluid side. Consequently, the evaluation of this term requires access to finite element information from both cells adjacent to an interface face. If the meshes on the two sides of the interface are locally nonmatching, meaning that an interface face on one side corresponds to several smaller faces on the other side due to local refinement, the interface integral cannot be evaluated directly on the full face. Instead, the integral is decomposed into contributions over the corresponding subfaces. This ensures that each contribution is evaluated on matching portions of the interface and that numerical quadrature is applied consistently, preserving the accuracy of the interface coupling.

Flux interpretation and sparsity pattern. From an algebraic point of view, the interface term differs fundamentally from the volume terms. While the latter only involve test and trial functions supported on the same cell, the interface contribution couples basis functions whose supports lie on *different cells* across Σ . In particular, displacement test functions supported on solid cells are coupled with velocity and pressure trial functions supported on neighboring fluid cells through an integral over their common face.

In this sense, the interface contribution plays the role of a numerical flux, in close analogy with discontinuous Galerkin formulations. As for DG flux terms, this coupling generates matrix entries connecting degrees of freedom belonging to distinct cells and therefore cannot be represented by a purely cell-wise sparsity pattern.

A fully flux-based sparsity pattern that couples all degrees of freedom on adjacent cells would certainly include all interface couplings, but it would also introduce many spurious entries corresponding to interactions between two fluid cells or two solid cells, which never arise in the bilinear form. This overestimation becomes particularly severe for higher-order elements, where the number of unnecessary matrix entries grows rapidly.

To balance correctness and efficiency, a restricted flux-based sparsity pattern is employed. On cells, the coupling structure reflects the underlying physical models: velocity components couple with each other and with the pressure in the fluid, while all displacement components are mutually coupled in the solid. Across interface faces, only couplings between displacement test functions on the solid side and velocity and pressure trial functions on the fluid side are admitted. While this component-wise restriction slightly overestimates the true set of interface couplings, it captures all physically relevant interactions while avoiding the excessive fill-in associated with a fully unrestricted flux sparsity pattern.

6.6 Velocity boundary conditions on the interface

A second difficulty in the implementation concerns the enforcement of velocity boundary conditions on the interior interface. While prescribing a zero velocity or a prescribed traction on the external boundary is straightforward, the coupled formulation also requires the fluid velocity to vanish on the interface,

$$u_h = 0 \quad \text{on } \Sigma,$$

reflecting the no-slip condition at the fluid-solid interface.

Unlike external boundaries, the interface Σ is an interior set of faces and does not carry a boundary indicator. Consequently, standard mechanisms for imposing Dirichlet boundary conditions cannot be applied directly. Instead, the constraint must be enforced by explicitly identifying all velocity degrees of freedom associated with interface faces and constraining them to zero.

The procedure follows a geometric inspection of the mesh. One loops over all active cells and considers only those belonging to the fluid subdomain. For each such cell, all interior faces are examined, and a face is identified as an interface face if the neighboring cell across that face belongs to the solid subdomain. In this case, all velocity

degrees of freedom supported on that face are constrained to zero. Pressure degrees of freedom are excluded from this process.

Special care is required when the neighboring solid cell is locally refined. In this situation, the fluid cell face may be adjacent to several smaller faces belonging to child cells on the solid side. The face is then classified as part of the interface if at least one of the neighboring child cells belongs to the solid subdomain. The velocity constraints are applied to the entire fluid face, ensuring that the no-slip condition is enforced consistently even in the presence of nonmatching meshes.

From an algebraic point of view, the constraint is imposed by introducing homogeneous linear constraints for the corresponding velocity degrees of freedom. Each constrained degree of freedom is fixed to zero, without coupling it to any other degree of freedom. This is achieved by defining a constraint equation of the form

$$u_i = 0,$$

which is sufficient for enforcing the desired boundary condition.

It is worth noting that this strategy relies on the assumption that the interface Σ coincides with a union of complete faces of coarse mesh cells. Under this assumption, once a fluid cell face is adjacent to a solid cell, all subfaces on the solid side belong to the solid subdomain as well. If this assumption were violated, it would be possible for only part of a face to lie on the interface, in which case the procedure would incorrectly constrain velocity degrees of freedom that do not belong to Σ . However, such configurations are excluded by the mesh assumptions adopted in this work.

7 Preconditioning

7.1 Linear solvers and block preconditioning

The discrete system derived in the previous section has a natural 3×3 block structure corresponding to the fluid velocity, fluid pressure, and solid displacement unknowns. While this formulation is convenient at the variational and assembly level, the design of scalable linear solvers benefits from reorganizing the algebraic structure of the problem.

Block structure induced by one-way coupling. The coupling between fluid and solid is one-way: the fluid exerts forces on the solid through the interface, while the solid displacement does not influence the fluid equations. As a consequence, after a suitable reordering of the degrees of freedom—placing all fluid velocity and pressure unknowns first, followed by the solid displacement unknowns—the global system matrix can be viewed conceptually as a 2×2 block matrix of the form

$$\mathbf{A}_{\text{global}} = \begin{bmatrix} \mathbf{A}_{\text{fluid}} & \mathbf{0} \\ \mathbf{B}_{\text{fs}} & \mathbf{A}_{\text{solid}} \end{bmatrix}.$$

Here, $\mathbf{A}_{\text{fluid}}$ denotes the Stokes operator acting on velocity and pressure, $\mathbf{A}_{\text{solid}}$ is the elasticity operator for the displacement, and $\mathbf{B}_{\text{fs}}$ collects the interface coupling terms.

It is important to emphasize that this 2×2 representation is a *conceptual reorganization*: in the implementation, the system is assembled and stored in the original 3×3 block form. Nevertheless, this viewpoint is essential for understanding the solver and preconditioner design.

Exact block factorization. Due to the block-triangular structure induced by the one-way coupling, the inverse of $\mathbf{A}_{\text{global}}$ can be written explicitly as

$$\mathbf{A}_{\text{global}}^{-1} = \begin{bmatrix} \mathbf{A}_{\text{fluid}}^{-1} & \mathbf{0} \\ -\mathbf{A}_{\text{solid}}^{-1} \mathbf{B}_{\text{fs}} \mathbf{A}_{\text{fluid}}^{-1} & \mathbf{A}_{\text{solid}}^{-1} \end{bmatrix}.$$

Applying this inverse to a vector $(x, y)^T$ requires one solve with $\mathbf{A}_{\text{fluid}}$ and one solve with $\mathbf{A}_{\text{solid}}$:

$$p_x = \mathbf{A}_{\text{fluid}}^{-1} x, \quad p_y = \mathbf{A}_{\text{solid}}^{-1} (y - \mathbf{B}_{\text{fs}} p_x).$$

This factorization reveals that the global problem can be solved by sequentially addressing the fluid and solid subproblems.

Preconditioning strategy. In practice, the exact inverses $\mathbf{A}_{\text{fluid}}^{-1}$ and $\mathbf{A}_{\text{solid}}^{-1}$ are not computed. Instead, they are replaced by efficient approximations

$$\tilde{\mathbf{A}}_{\text{fluid}}^{-1} \approx \mathbf{A}_{\text{fluid}}^{-1}, \quad \tilde{\mathbf{A}}_{\text{solid}}^{-1} \approx \mathbf{A}_{\text{solid}}^{-1},$$

leading to the block preconditioner

$$\tilde{\mathbf{A}}_{\text{global}}^{-1} = \begin{bmatrix} \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \mathbf{0} \\ -\tilde{\mathbf{A}}_{\text{solid}}^{-1} \mathbf{B}_{\text{fs}} \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \tilde{\mathbf{A}}_{\text{solid}}^{-1} \end{bmatrix}.$$

This construction shows explicitly that the objects appearing in place of the inverses are themselves *preconditioners*. The global preconditioner therefore reduces the coupled problem to the design of robust and scalable preconditioners for the fluid and solid subproblems separately.

Recursive structure of the fluid block. The fluid operator $\mathbf{A}_{\text{fluid}}$ still possesses an internal saddle-point structure due to the coupling between velocity and pressure. Consequently, $\tilde{\mathbf{A}}_{\text{fluid}}^{-1}$ is itself constructed using a block preconditioning strategy based on a Schur complement for the pressure. In this sense, the overall solver employs a *recursive block structure*: a block preconditioner for the coupled fluid–solid system, within which the fluid block is treated by another block preconditioner tailored to the Stokes equations.

Preconditioning of the Stokes subproblem The fluid block $\mathbf{A}_{\text{fluid}}$ corresponds to a discretization of the Stokes operator and exhibits the classical saddle-point structure associated with the coupling between velocity and pressure. In block form, it can be written as

$$\mathbf{A}_{\text{fluid}} = \begin{bmatrix} \mathbf{A}_{\text{stokes}} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{0} \end{bmatrix},$$

where $\mathbf{A}_{\text{stokes}}$ denotes the discrete viscous operator acting on the velocity unknowns and $\mathbf{B}$ represents the discrete divergence operator.

Following a standard block preconditioning approach for Stokes problems, we employ the approximate inverse

$$\tilde{\mathbf{A}}_{\text{fluid}}^{-1} = \begin{bmatrix} \mathbf{A}_{\text{stokes}}^{-1} & \mathbf{0} \\ \mathbf{B} & \nu^{-1} \mathbf{M}_p^{-1} \end{bmatrix},$$

where $\mathbf{M}_p$ is the pressure mass matrix and ν is the kinematic viscosity. This choice is motivated by the fact that the Schur complement associated with the pressure is spectrally equivalent to $\nu^{-1} \mathbf{M}_p$, yielding a robust and scalable preconditioner for the Stokes system.

Global block preconditioner Returning to the original ordering of the unknowns, the fully coupled discrete system exhibits a 3×3 block structure corresponding to fluid velocity, fluid pressure, and solid displacement. The global block preconditioner is constructed accordingly as

$$\tilde{\mathbf{A}}_{\text{global}}^{-1} = \begin{bmatrix} \mathbf{A}_{\text{stokes}}^{-1} & \mathbf{0} & \mathbf{0} \\ \mathbf{B} & \nu^{-1} \mathbf{M}_p^{-1} & \mathbf{0} \\ -\mathbf{P}_{\text{sol}}^{-1} \mathbf{D}_u \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & -\mathbf{P}_{\text{sol}}^{-1} \mathbf{D}_p \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \mathbf{P}_{\text{sol}}^{-1} \end{bmatrix}.$$

Here, $\mathbf{P}_{\text{sol}}^{-1}$ denotes a preconditioner for the solid elasticity operator, while the matrices $\mathbf{D}_u$ and $\mathbf{D}_p$ represent the coupling operators between the fluid velocity and pressure with the solid displacement. For convenience, we introduce the compact notation

$$\mathbf{D} := [\mathbf{D}_u \quad \mathbf{D}_p].$$

Applying the global preconditioner to a block vector yields

$$\begin{bmatrix} \mathbf{y}_u \\ \mathbf{y}_p \\ \mathbf{y}_s \end{bmatrix} = \tilde{\mathbf{A}}_{\text{global}}^{-1} \begin{bmatrix} \mathbf{x}_u \\ \mathbf{x}_p \\ \mathbf{x}_s \end{bmatrix},$$

with the individual components computed sequentially as

$$\mathbf{y}_u = \mathbf{A}_{\text{stokes}}^{-1} \mathbf{x}_u,$$

$$\mathbf{y}_p = \nu^{-1} \mathbf{M}_p^{-1} (\mathbf{x}_p - \mathbf{B} \mathbf{y}_u),$$

and

$$\mathbf{y}_s = \mathbf{P}_{\text{sol}}^{-1} (\mathbf{x}_s - \mathbf{D}_u \mathbf{y}_u - \mathbf{D}_p \mathbf{y}_p) = \mathbf{P}_{\text{sol}}^{-1} \left(\mathbf{x}_s - \mathbf{D} \begin{bmatrix} \mathbf{y}_u \\ \mathbf{y}_p \end{bmatrix} \right).$$

This formulation makes explicit the lower-triangular structure of the preconditioner: the fluid velocity is preconditioned first, followed by a pressure correction that accounts for the velocity contribution, and finally the solid right-hand side is corrected using both fluid components before applying the solid preconditioner.

Balancing sub-preconditioners. An important practical consideration is that the performance of the global preconditioner is not determined solely by the quality of the individual sub-preconditioners in isolation. When combined into a multiphysics block preconditioner, the fluid and solid preconditioners should exhibit comparable convergence behavior. A very inexpensive but weak preconditioner for one subproblem may increase the overall iteration count, while an overly expensive preconditioner for the other may dominate the cost per iteration. Achieving an appropriate balance between these effects is therefore essential for overall efficiency.

The specific choices of preconditioners for the fluid and solid subproblems are discussed in the following subsection.

7.2 Sub-preconditioners

To efficiently solve the monolithic linear system arising from the discretization, we employ a block-triangular preconditioner. The global system is decomposed into fluid velocity ($A_{\mathbf{uu}}$), fluid pressure (M_p), and solid displacement ($A_{\mathbf{ss}}$) blocks. The diagonal blocks are inverted approximately using Algebraic Multigrid (AMG) methods provided by the Trilinos ML package.

For the vector-valued blocks—specifically the fluid velocity and solid displacement—standard scalar AMG aggregation is insufficient due to the coupling of vector components. To address this, we explicitly construct the near null-space (constant modes) of the underlying finite element space and supply it to the AMG algorithm. This ensures that rigid body modes are preserved on coarser grids. The pressure block is preconditioned using the pressure mass matrix (M_p), which serves as a robust approximation for the pressure Schur complement in incompressible flow.

The AMG hierarchy is configured with specific parameters to maximize robustness for the coupled physics. The `AdditionalData` structure is initialized as follows:

- **Elliptic (true):** This flag informs the solver that the underlying operators ($A_{\mathbf{uu}}$ and $A_{\mathbf{ss}}$) are symmetric and positive definite (SPD), allowing the use of symmetric smoothers.
- **Higher Order Elements (true):** This adjusts the aggregation algorithm to account for the denser matrix connectivity typical of higher-order finite element discretizations or vector-valued problems, ensuring that node aggregation remains effective.
- **Smoother Sweeps (2):** We perform two sweeps of the smoother (Symmetric Gauss-Seidel) at each multigrid level to improve the damping of high-frequency error components.
- **Aggregation Threshold (0.02):** Weak connections in the matrix graph (below 2% of the maximum entry) are dropped during the aggregation phase to prevent the formation of poor coarse-grid operators.

The global preconditioner is applied via a block-Gauss-Seidel procedure with an inner Conjugate Gradient (CG) solver for each block, solved to a relative tolerance of 10^{-2} .

8 Code Verification

8.1 Manufactured solution

To verify our code, we constructed a manufactured solution that allows us to check both the convergence of the solver and its expected order of accuracy. Due to the complex shape of the interface Σ in our domain, we employed empirical considerations to define a solution that satisfies all necessary constraints in a straightforward manner. The construction procedure is implemented via a Mathematica script in the `scripts` folder of our repository, automating the calculations and minimizing the risk of errors.

The first condition to satisfy is the incompressibility of the fluid:

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega_{\text{fluid}}.$$

This is ensured by defining a sufficiently smooth (at least C^2) stream function ψ , and setting the velocity components as

$$u_1 = \frac{\partial \psi}{\partial y}, \quad u_2 = -\frac{\partial \psi}{\partial x}.$$

By the Schwarz theorem, this guarantees that $\nabla \cdot \mathbf{u} = 0$.

To enforce the no-slip condition on the interface Σ , we multiply the velocity field by a function that vanishes on Σ . Squaring this function ensures that the derivatives of $\mathbf{u}$, which enter the stress tensor, also vanish on Σ . In our implementation, the zero function on Σ is constructed by multiplying each component of the velocity by a factor that is zero along the discrete points of Σ .

This procedure yields a velocity field $\mathbf{u}$ that satisfies the incompressibility constraint and the Dirichlet conditions on the interface. The corresponding Dirichlet and Neumann boundary conditions, as well as the forcing terms in the momentum and elasticity equations, can then be derived accordingly.

The second consideration arises from the fluid-solid coupling condition on the interface:

$$\sigma(\mathbf{d}) \mathbf{n} = (2\nu \varepsilon(\mathbf{u}) - p \mathbf{I}) \mathbf{n}, \quad \text{on } \Sigma. \quad (69)$$

Finding an analytical expression that satisfies this condition for a discontinuous normal vector $\mathbf{n}$ is challenging. We therefore observed the structure of the solid and fluid stress tensors:

$$\sigma(\mathbf{d}) = 2\mu \varepsilon(\mathbf{d}) + \lambda (\text{tr } \varepsilon(\mathbf{d})) \mathbf{I}, \quad \sigma_{\text{fluid}} = 2\nu \varepsilon(\mathbf{u}) - p \mathbf{I}.$$

Exploiting the similarity between these tensors, we can simplify the construction of the manufactured solution. For example, setting $\mathbf{d} = \mathbf{u}$ (or a suitable fraction of $\mathbf{u}$), choosing $\nu = \mu$, and defining the pressure as

$$p = -\lambda \text{tr}(\varepsilon(\mathbf{d}))$$

ensures that the stress tensors match pointwise, independently of the local normal vector.

Since $\mathbf{d}$ is proportional to $\mathbf{u}$, it is divergence-free as well, implying $\text{tr}(\varepsilon(\mathbf{d})) = 0$ on Σ . Hence, we set the pressure $p = 0$ on Σ and include the same zero function used for the velocity field.

With these choices for $\mathbf{u}$, $\mathbf{d}$, and p that satisfy the essential constraints of the problem, we then construct the corresponding forcing terms for the momentum and elasticity equations. The Dirichlet boundary conditions are enforced by assigning the exact solution on the boundary, while the Neumann conditions are derived by evaluating the corresponding stress tensors at the boundaries.

This procedure ensures that our manufactured solution satisfies all required interface and boundary conditions and provides a robust test for verifying convergence and accuracy of our solver.

At the moment, the implementation of the manufactured solution is still in progress. While the methodology has been fully developed and automated through the Mathematica script, the solver has not yet been fully tested with this manufactured solution. Completion of this implementation will allow a comprehensive verification of the code, including convergence studies and validation of the expected order of accuracy.

8.2 Verification Using Refined Solver Solution

To further validate our implementation under the actual boundary data, and to ensure that the solution converges not only for the constructed manufactured solution but also in a more realistic setting, we employed an alternative approach. Specifically, we ran our solver on a highly refined mesh and used the resulting solution as a reference. Under the assumption that the solver is convergent, this reference solution can be considered highly accurate and thus suitable for testing the correctness of our implementation.

While this approach does not guarantee that the solver will always converge to the exact solution, it provides a reliable benchmark to assess the convergence behavior and verify the implementation when the solver approaches the correct solution. At the moment, also this implementation is still in progress

9 Results

9.1 Strong scaling

We investigated the parallel performance of the proposed solver from a simulation point of view, with a particular focus on its strong scaling behavior. The analysis was carried out by measuring the computational time required to solve a fixed problem while varying the number of processors. In particular, we monitored the time spent in the main computational phases, namely the system assembly, the linear solver, and the total execution time.

The results, reported in the following figures, show the expected trend of decreasing execution time as the number of processing cores increases. These results can be compared with the ideal strong scaling behavior, which assumes a perfectly parallelizable workload.

All simulations were performed on a high-performance computing cluster, and results are reported for configurations using up to 32 cores. Improved parallel efficiency is observed in the three-dimensional case, where the larger number of degrees of freedom provides a more favorable ratio between parallel and serial workloads.

Regarding the influence of mesh size on parallel scaling, the observed behavior is consistent with Amdahl's law. For very coarse meshes, the parallel efficiency is limited, as the serial portion of the computation dominates the total execution time. As the mesh is refined, parallel scalability improves due to the increased computational workload per processor. However, beyond a certain point, scalability tends to stabilize and may eventually deteriorate as the computation becomes memory-bound.

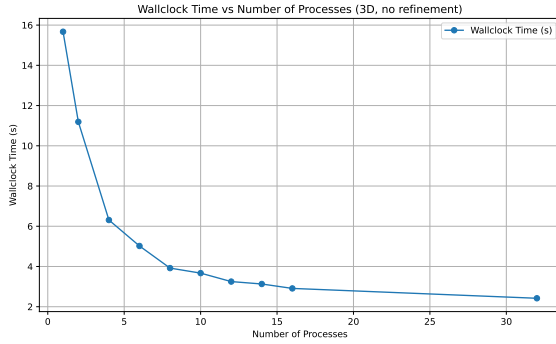


Figure 3: Execution time for 3D mesh.

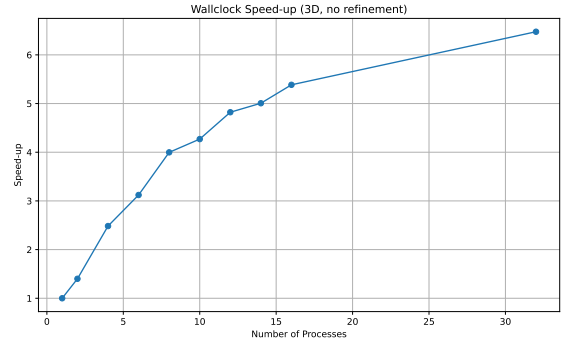


Figure 4: Speedup for 3D mesh.

9.2 Iterative vs Direct

We have also tested and compared our implemented iterative solver against a direct solver. For small-scale 2D problems, the direct solver outperforms our iterative implementation due to the low computational overhead. However, as the mesh is refined and the problem size increases to levels typical of realistic 3D simulations, the limitations of the direct solver become apparent. In particular, direct methods exhibit significantly higher memory consumption and longer computational times, which can make them impractical for large-scale problems.

As expected, for sufficiently fine 3D meshes, our iterative solver clearly outperforms the direct method, both in terms of memory efficiency and execution time, demonstrating its suitability for large-scale simulations and more realistic problem settings.

10 Discussion

The implemented iterative solver has been validated and performs correctly across all tested scenarios. In comparison with the direct solver, the iterative approach demonstrates superior scalability, particularly for large and refined meshes, where direct methods become increasingly memory-intensive and computationally expensive.

The use of adaptive mesh refinement provides a significant reduction in computational time while maintaining solution accuracy. Thanks to the carefully designed preconditioner, the number of outer iterations remains essentially constant, independently of the mesh refinement level. This robustness is observed consistently in both 2D and 3D simulations.

Overall, the combination of an efficient iterative solver, an optimal preconditioner, and adaptive refinement allows for a highly effective solution strategy that is well-suited for large-scale and realistic fluid–solid interaction problems. The approach ensures both computational efficiency and numerical stability, making it preferable to direct solvers in practical scenarios.

11 Conclusions

We have developed an efficient and scalable solver for the coupled fluid–solid problem, which demonstrates good parallel performance even on large-scale simulations.

The analysis of the computational results revealed several important insights. As expected, the solution phase dominates the overall computational cost. However, the other parts of the simulation also consume more time than initially anticipated. This suggests that further optimizations could be explored, potentially leveraging task-based parallelism and the workstream mechanisms provided by `deal.II`.

Regarding future developments, several directions are possible:

- Completion of one of the ongoing validation cases to fully assess solver accuracy.
- Implementation of a matrix-free approach combined with a multigrid method capable of efficiently handling the saddle-point structure of the Stokes problem.

Overall, the solver provides a robust and scalable foundation for large-scale simulations, and the identified opportunities for optimization and extension offer promising directions for further improvement.

12 Appendix

12.1

Green's formula for second-order tensors The Green (integration by parts) formula for a second-order tensor field $\mathbf{T}$ and a vector-valued test function $\mathbf{v}$ reads

$$\int_{\Omega} (\nabla \cdot \mathbf{T}) \cdot \mathbf{v} \, d\Omega = \int_{\partial\Omega} (\mathbf{T}\mathbf{n}) \cdot \mathbf{v} \, d\Gamma - \int_{\Omega} \mathbf{T} : \nabla \mathbf{v} \, d\Omega. \quad (70)$$

This identity generalizes the classical divergence theorem to tensor-valued fields and plays a fundamental role in the derivation of weak formulations for problems involving stresses or fluxes, such as the Stokes and linear elasticity equations.

12.2

Integration by parts for the symmetric gradient

We recall the integration by parts formula associated with the symmetric gradient operator. For sufficiently smooth vector fields $\mathbf{u}$ and $\mathbf{v}$, it holds that

$$\int_{\Omega} (\nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u})) \cdot \mathbf{v} \, d\Omega = \int_{\partial\Omega} (\boldsymbol{\varepsilon}(\mathbf{u})\mathbf{n}) \cdot \mathbf{v} \, d\Gamma - \int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega, \quad (71)$$

where $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ denotes the symmetric gradient and $\mathbf{n}$ is the outward unit normal to $\partial\Omega$.

This identity can be interpreted as a generalization of the classical divergence (Gauss) theorem to symmetric tensor fields and plays a fundamental role in the derivation of the weak formulation of the Stokes equations. In particular, it shows that the viscous term naturally gives rise to a bilinear and symmetric form involving the symmetric gradient. Under suitable boundary conditions, this formulation yields a positive definite contribution, which is essential for the stability and well-posedness of the finite element discretization.

12.3

Lemma 1 (Lax–Milgram). Let V be a Hilbert space, $a(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$ a continuous and coercive bilinear form, and $F(\cdot) : V \rightarrow \mathbb{R}$ a linear and continuous functional. Then, there exists a unique solution to the problem

$$\text{find } u \in V : \quad a(u, v) = F(v) \quad \forall v \in V. \quad (72)$$

12.4

Corollary of Lax–Milgram The solution of equation 72 is bounded by the data, that is

$$\|u\|_V \leq \frac{1}{\alpha} \|F\|_{V'},$$

where α is the coercivity constant of the bilinear form $a(\cdot, \cdot)$, while $\|F\|_{V'}$ is the norm of the functional F .

12.5

Theorem 1 (F. Brezzi). The Galerkin approximation for the problem of Stokes admits one and only one solution if the following conditions hold:

1. The bilinear form $a(\cdot, \cdot)$ is:

- (a) coercive, that is $\exists \alpha > 0$ (possibly depending on h) such that

$$a(v_h, v_h) \geq \alpha \|v_h\|_V^2 \quad \forall v_h \in V_h^*,$$

where

$$V_h^* = \{v_h \in V_h : b(v_h, q_h) = 0 \, \forall q_h \in Q_h\};$$

(b) continuous, that is $\exists \gamma > 0$ such that

$$|a(u_h, v_h)| \leq \gamma \|u_h\|_V \|v_h\|_V \quad \forall u_h, v_h \in V_h.$$

2. The bilinear form $b(\cdot, \cdot)$ is continuous, that is $\exists \delta > 0$ such that

$$|b(v_h, q_h)| \leq \delta \|v_h\|_V \|q_h\|_Q \quad \forall v_h \in V_h, q_h \in Q_h.$$

3. Finally, there exists a positive constant β (possibly depending on h) such that

$$\forall q_h \in Q_h, \exists v_h \in V_h : \quad b(v_h, q_h) \geq \beta \|v_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}.$$

Under the previous assumptions the discrete solution fulfills the following a-priori estimates:

$$\|u_h\|_V \leq \frac{1}{\alpha} \|f\|_{V'},$$

$$\|p_h\|_Q \leq \frac{1}{\beta} \left(1 + \frac{\gamma}{\alpha}\right) \|f\|_{V'},$$

where V' is the dual space of V . Moreover, the following convergence results hold:

$$\|u - u_h\|_V \leq \left(1 + \frac{\delta}{\beta}\right) \left(1 + \frac{\gamma}{\alpha}\right) \inf_{v_h \in V_h} \|u - v_h\|_V + \frac{\delta}{\alpha} \inf_{q_h \in Q_h} \|p - q_h\|_Q,$$

$$\|p - p_h\|_Q \leq \frac{\gamma}{\beta} \left(1 + \frac{\gamma}{\alpha}\right) \left(1 + \frac{\delta}{\beta}\right) \inf_{v_h \in V_h} \|u - v_h\|_V + \left(1 + \frac{\delta}{\beta} + \frac{\delta\gamma}{\alpha\beta}\right) \inf_{q_h \in Q_h} \|p - q_h\|_Q.$$

12.6

Korn's inequality. Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, and let

$$V := \{\mathbf{v} \in [H^1(\Omega)]^d : \mathbf{v} = 0 \text{ on } \Gamma_D\},$$

where $\Gamma_D \subset \partial\Omega$ has positive measure. Then there exists a constant $C_K > 0$, depending only on Ω and Γ_D , such that

$$\|\nabla \mathbf{v}\|_{L^2(\Omega)} \leq C_K \|\boldsymbol{\varepsilon}(\mathbf{v})\|_{L^2(\Omega)} \quad \forall \mathbf{v} \in V,$$

where $\boldsymbol{\varepsilon}(\mathbf{v}) := \frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^\top)$ denotes the symmetric gradient.

This inequality implies that the seminorm $\|\nabla \mathbf{v}\|_{L^2(\Omega)}$ is equivalent to the norm $\|\mathbf{v}\|_{H^1(\Omega)}$ on V .

12.7

Trace inequality Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain with boundary $\partial\Omega$, and let $V = [H^1(\Omega)]^d$. Then, there exists a constant $C_{\text{tr}} > 0$, depending only on Ω , such that for all $\mathbf{v} \in V$ the following inequality holds:

$$\|\mathbf{v}\|_{H^{1/2}(\partial\Omega)} \leq C_{\text{tr}} \|\mathbf{v}\|_{H^1(\Omega)}. \quad (73)$$

Equivalently, for a Lipschitz subset $\Sigma \subset \partial\Omega$, we have

$$\|\mathbf{v}\|_{H^{1/2}(\Sigma)} \leq C_{\text{tr}} \|\mathbf{v}\|_{H^1(\Omega)}, \quad (74)$$

which allows to control the boundary (trace) norm by the volumetric H^1 -norm.

This result justifies the continuity of linear forms defined on the boundary, such as

$$d(\mathbf{u}, p; \mathbf{v}) = \int_{\Sigma} \mathbf{t} \cdot \mathbf{v} \, d\Gamma,$$

provided that $\mathbf{t} \in [H^{-1/2}(\Sigma)]^d$.

12.8

Lemma (Céa). Let V be a Hilbert space with norm $\|\cdot\|_V$, and let $a(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$ be a continuous and coercive bilinear form, i.e., there exist constants $\alpha, \gamma > 0$ such that

$$a(v, v) \geq \alpha \|v\|_V^2 \quad \forall v \in V, \quad |a(u, v)| \leq \gamma \|u\|_V \|v\|_V \quad \forall u, v \in V.$$

Let $u \in V$ be the unique solution of

$$a(u, v) = F(v) \quad \forall v \in V,$$

for a continuous linear functional $F \in V'$, and let $V_h \subset V$ be a finite-dimensional subspace. Denote by $u_h \in V_h$ the Galerkin approximation solving

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h.$$

Then, the following estimate holds:

$$\|u - u_h\|_V \leq \frac{\gamma}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (75)$$

Remark. This lemma shows that the error of the Galerkin approximation is bounded by the best approximation error in the chosen finite-dimensional space. It is a fundamental tool for deriving convergence estimates in finite element analysis.

References